Lab 3 — Problem Statement #55

LAB 3 — COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Problem Statement #55 — Book Club Reading Challenge & Discussion Portal

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1. Architectural Pattern Selection

We chose Layered Architecture for the Book Club Reading Challenge & Discussion Portal System.

Selected Architecture: Layered Architecture

Justification: Layered Architecture is suitable for the Book Club Reading Challenge & Discussion Portal because the system has clear presentation, business and data responsibilities. The Club Member UI handles interaction, the application components handle reading goals, discussions, spoiler control, voting and authentication, and the database handles persistent data.

Reason 1 — Separation of concerns: The portal's user interface, business rules and data storage can be kept separate. This makes the reading-goal, discussion, spoiler and voting functions easier to develop, test and maintain.

Reason 2 — Appropriate complexity: Problem Statement #55 describes a focused community portal. A microservices design would add unnecessary deployment, monitoring, communication and data-consistency overhead. Layered Architecture provides the required modularity with a simpler design.

Security advantage: Authentication and session verification can be applied before protected operations such as voting and access to protected discussion/member information. This supports the stated authentication and authorization requirements.

Performance benefit: The application can perform targeted database operations for reading goals, discussions and polls. A controlled vote-check and update flow also supports timely poll-tally updates and avoids unnecessary processing.

2. Component Identification

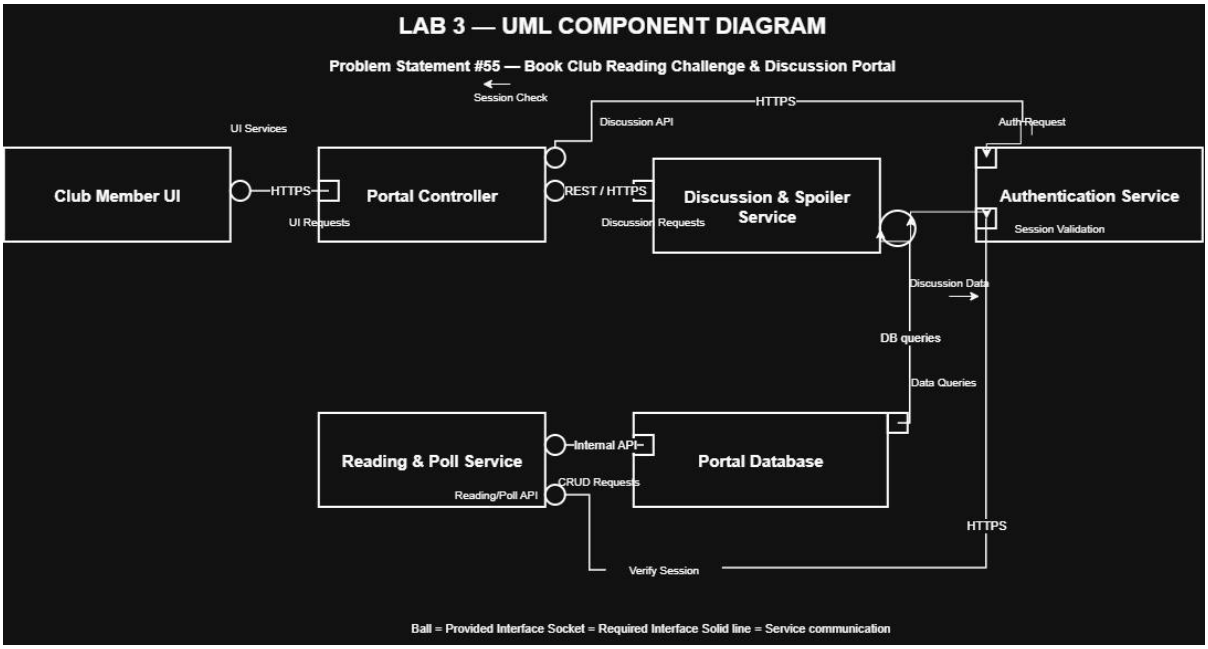
Component	Responsibility
Club Member UI	Provides the interface for reading goals, chapter discussions, spoiler reveal and monthly voting.
Portal Controller	Receives requests and coordinates the required application services.
Discussion & Spoiler Service	Manages chapter discussions/replies and controls spoiler visibility.
Reading & Poll Service	Manages reading goals, monthly polls, voting, duplicate-vote checking and poll tally.
Authentication Service	Verifies the member session before protected operations.
Portal Database	Stores member, book/chapter, reading-goal, discussion, poll and vote information.

3. Interfaces

Interface	Connected Components	Purpose
UI Services / UI Requests	Club Member UI ↔ Portal Controller	Transfers member requests and responses.
Discussion API / Discussion Requests	Portal Controller ↔ Discussion & Spoiler Service	Creates and accesses chapter discussions and spoiler-controlled content.
Reading/Poll API / Service Calls	Portal Controller ↔ Reading & Poll Service	Handles reading goals, polls and voting.
Discussion Data / Data Queries	Discussion & Spoiler Service ↔ Portal Database	Reads and stores discussion data.
Internal API / CRUD Requests	Reading & Poll Service ↔ Portal Database	Stores and retrieves goals, polls and votes.
Session Check / Auth Request	Portal Controller ↔ Authentication Service	Checks whether the member session is valid.
Verify Session / Session Validation	Reading & Poll Service ↔ Authentication Service	Validates the session before protected voting operations.

4. UML Component Diagram

The diagram contains six components, exceeding the required five, and seven interface interactions, exceeding the required four.



5. Conclusion

Layered Architecture is selected because it provides clear separation of responsibilities, supports the authentication and voting requirements, and is appropriate for the size and complexity of the Book Club Reading Challenge & Discussion Portal.